Q1.

For this project, I selected Cryptochrome-1 (CRY1) from Homo sapiens as my starting protein.

The RefSeq accession number for this protein is NP_004066.1, or Q16526 for UniProt.

Q2.

Among the top hits, a strong match was identified as **Bos taurus cDNA clone LB01721.CR_O20 GC_BGC-17 IMAGE:8095822 5′, mRNA sequence** (GenBank accession DV817148.1). The sequence was annotated only as a cDNA clone with no defined functional name, making it a suitable candidate for a potentially novel homolog of human CRY1.

The alignment between the human CRY1 query and the Bos taurus sequence showed **an E-value of 0.0, a bit score of 609, and 99.3% sequence identity across a 901-nt region**. These results indicate a very high level of homology, suggesting that this Bos taurus clone encodes a protein closely related to human Cryptochrome-1.

LB01721.CR_O20 GC_BGC-17 Bos taurus cDNA clone IMAGE:8095822 5', mRNA sequence

Sequence ID: DV817148.1 Length: 901 Number of Matches: 1

Range 1: 8 to 901 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
609 bits(1571)	0.0	Compositional matrix adjust.	296/298(99%)	297/298(99%)	0/298(0%)	+2

```
Query  1    MGVNAVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINRWRFLLQCLE  60
            MGVN+VHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINRWRFLLQCLE
Sbjct  8    MGVNSVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINRWRFLLQCLE  187

Query  61   DLDANLRKLNSRLFVIRGQPADVFPRLFKEWNITKLSIEYDSEPFGKERDAAIKKLATEA  120
            DLDANLRKLNSRLFVIRGQPADVFPRLFKEWNITKLSIEYDSEPFGKERDAAIKKLATEA
Sbjct  188  DLDANLRKLNSRLFVIRGQPADVFPRLFKEWNITKLSIEYDSEPFGKERDAAIKKLATEA  367

Query  121  GVEVIVRISHTLYDLDKIIELNGGQPPLTYKRFQTLISKMEPLEIPVETITSEVIEKCTT  180
            GVEVIVRISHTLYDLDKIIELNGGQPPLTYKRFQTLISKMEPLEIPVETITSEVIEKCTT
Sbjct  368  GVEVIVRISHTLYDLDKIIELNGGQPPLTYKRFQTLISKMEPLEIPVETITSEVIEKCTT  547

Query  181  PLSDDHDEKYGVPSLEELGFDTDGLSSAVWPGGETEALTRLERHLERKAWVANFERPRMN  240
            PLSDDHDEKYGVPSLEELGFDTDGL SAVWPGGETEALTRLERHLERKAWVANFERPRMN
Sbjct  548  PLSDDHDEKYGVPSLEELGFDTDGLPSAVWPGGETEALTRLERHLERKAWVANFERPRMN  727

Query  241  ANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLYKKVKKNSSPPLSLYGQLLWREFFYT  298
            ANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLYKKVKKNSSPPLSLYGQLLWREFFYT
Sbjct  728  ANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLYKKVKKNSSPPLSLYGQLLWREFFYT  901
```

Q3.

Protein sequence:

```
>putative_CRY1_like|Bos_taurus|DV817148.1|ORF2|frame+2|len=297aa

MGVNSVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGIN

RWRFLLQCLEDLDANLRKLNSRLFVIRGQPADVFPRLFKEWNITKLSIEY

DSEPFGKERDAAIKKLATEAGVEVIVRISHTLYDLDKIIELNGGQPPLTY
```

```
KRFQTLISKMEPLEIPVETITSEVIEKCTTPLSDDHDEKYGVPSLEELGF

DTDGLPSAVWPGGETEALTRLERHLERKAWVANFERPRMNANSLLASPTG

LSPYLRFGCLSCRLFYFKLTDLYKKVKKNSSPPLSLYGQLLWREFFYT
```

Name (provisional): Putative cryptochrome-1–like protein

Species: Bos taurus (from cDNA clone LB01721.CR_O20 GC_BGC-17 IMAGE:8095822; GenBank DV817148.1)

How obtained: The nucleotide sequence DV817148.1 was translated with NCBI ORFfinder (standard code; ATG start; six-frame translation). I selected the longest ORF (ORF2, + strand, frame +2, 894 nt / 297 aa) with no internal stops.

Q4.

As shown by figure below, none of my hits had 100% identity to this protein. The best BLASTP hit is Cryptochrome-1 from Bos mutus (ELR57829.1) with 99.66% identity, 0.0 E-value, and 0 gaps over 298 aa. Because the top match is <100% identical and from a different species than the source clone (Bos taurus), this satisfies the project's definition of a novel homolog.

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☑	Cryptochrome-1 [Bos mutus]	Bos mutus	615	615	100%	0.0	99.66%	589	ELR57829.1
☑	cryptochrome-1 isoform X2 [Callorhinus ursinus]	Callorhinus ursinus	614	614	100%	0.0	99.66%	611	XP_073756447.1
☑	cryptochrome-1 isoform X2 [Pteronotus mesoamericanus]	Pteronotus mesoamericanus	614	614	100%	0.0	99.66%	616	XP_054430553.1
☑	cryptochrome-1 isoform X4 [Vicugna pacos]	Vicugna pacos	614	614	100%	0.0	99.66%	627	XP_072829164.1
☑	cryptochrome-1 isoform X2 [Odocoileus virginianus]	Odocoileus virginianus	614	614	100%	0.0	99.66%	628	XP_070310107.1
☑	cryptochrome-1 isoform X1 [Capricornis sumatraensis]	Capricornis sumatraensis	614	614	100%	0.0	99.66%	628	XP_068826436.1
☑	cryptochrome-1 isoform X2 [Bubalus kerabau]	Bubalus kerabau	614	614	100%	0.0	99.66%	628	XP_055430804.1
☑	cryptochrome-1 isoform X2 [Bos taurus]	Bos taurus	614	614	100%	0.0	99.66%	628	XP_059742545.1
☑	cryptochrome-1 isoform X1 [Vicugna pacos]	Vicugna pacos	613	613	100%	0.0	99.66%	587	XP_006205926.1
☑	cryptochrome-1 isoform X1 [Perognathus longimembris pacificus]	Perognathus longimembris p...	613	613	100%	0.0	99.66%	587	XP_048212448.1
☑	cryptochrome-1 [Budorcas taxicolor]	Budorcas taxicolor	613	613	100%	0.0	99.66%	587	XP_052495686.1
☑	cryptochrome-1 isoform X2 [Capricornis sumatraensis]	Capricornis sumatraensis	613	613	100%	0.0	99.66%	627	XP_068826437.1
☑	cryptochrome-1 isoform X2 [Desmodus rotundus]	Desmodus rotundus	613	613	100%	0.0	99.66%	611	XP_053777980.1
☑	cryptochrome-1 isoform X2 [Dama dama]	Dama dama	613	613	100%	0.0	99.66%	628	XP_060981421.1

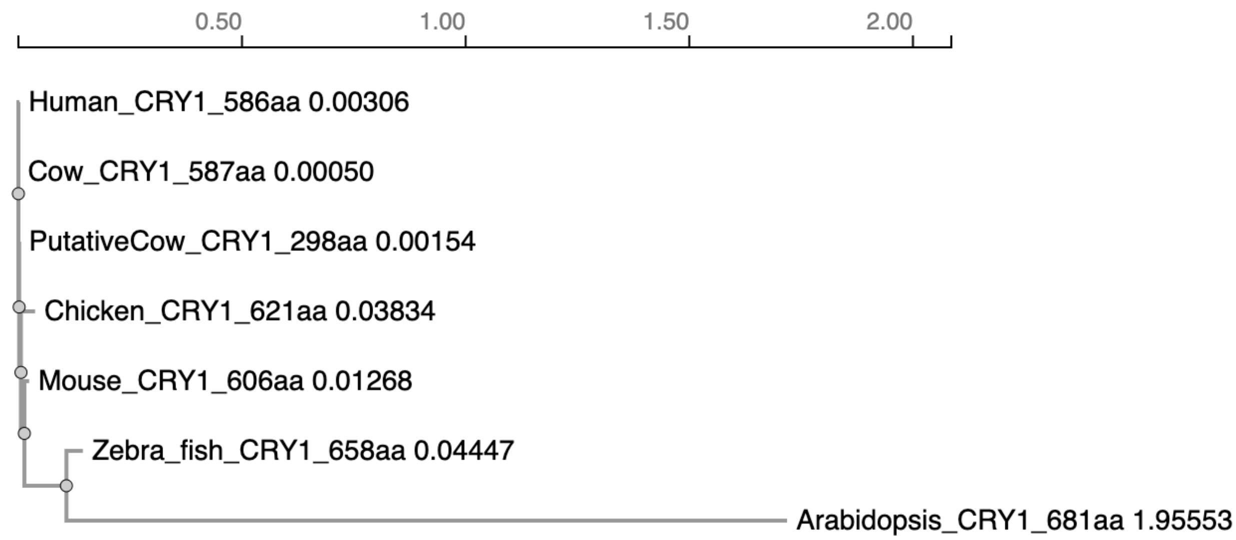
Q5.

The members I chose, aside from Cryptochrome-1 and Putative cryptochrome-1-like protein, are: Mus musculus CRY1, Bos taurus CRY1, Gallus gallus CRY1, Danio rerio CRY1a or CRY1b, and Arabidopsis thaliana CRY1.

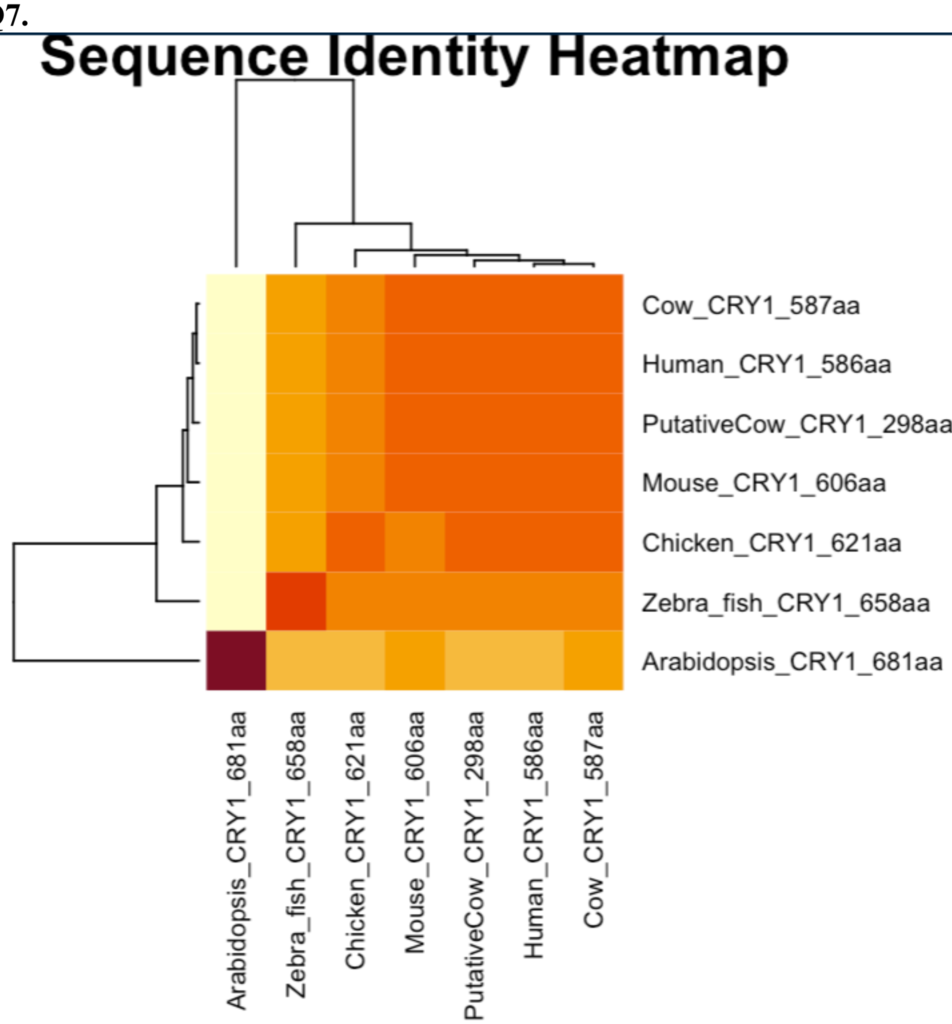
```
Human_CRY1_586aa        ---------MGVNAVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINR
Mouse_CRY1_606aa        ---------MGVNAVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINR
Cow_CRY1_587aa          ---------MGVNAVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINR
Chicken_CRY1_621aa      ---------MGVNAVHWFRKGLRLHDNPALRECIRGADTVRCVYILDPWFAGSSNVGINR
Zebra_fish_CRY1_658aa   ---------MAPNSIHWFRKGLRLHDNPALQEAVRGADTVRCVYFLDPWFAGSSNLGVNR
```

```
Arabidopsis_CRY1_681aa  MSGSVSGCGSGGCSIVWFRRDLRVEDNPALAAAVRAGPVI-ALFVWAPEEEGHYHPGRVS
PutativeCow_CRY1_298aa  ---------MGVNSVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINR

Human_CRY1_586aa        WRFLLQCLEDLDANLRKLNSRLFVIRGQPAD-VFPRLFKEWNITKLSIEYDSEPFGKERD
Mouse_CRY1_606aa        WRFLLQCLEDLDANLRKLNSRLFVIRGQPAD-VFPRLFKEWNITKLSIEYDSEPFGKERD
Cow_CRY1_587aa          WRFLLQCLEDLDANLRKLNSRLFVIRGQPAD-VFPRLFKEWNITKLSIEYDSEPFGKERD
Chicken_CRY1_621aa      WRFLLQCLEDLDANLRKLNSRLFVIRGQPAD-VFPRLFKEWSIAKLSIEYDSEPFGKERD
Zebra_fish_CRY1_658aa   WRFLLQCLDDLDSNLRKLNSRLFVVRGQPAN-VFPRLFKEWKISRLTFEYDSEPFGKERD
Arabidopsis_CRY1_681aa  RWWLKNSLAQLDSSLRSLGTCLITKRSTDSVASLLDVVKSTGASQIFFNHLYDPLSLVRD
PutativeCow_CRY1_298aa  WRFLLQCLEDLDANLRKLNSRLFVIRGQPAD-VFPRLFKEWNITKLSIEYDSEPFGKERD

Human_CRY1_586aa        AAIKKLATEAGVEVIVRISHTLYDLDKIIELNGGQPPLTYKRFQT----LI-SKMEPLEI
Mouse_CRY1_606aa        AAIKKLATEAGVEVIVRISHTLYDLDKIIELNGGQPPLTYKRFQT----LV-SKMEPLEM
Cow_CRY1_587aa          AAIKKLATEAGVEVIVRISHTLYDLDKIIELNGGQPPLTYKRFQT----LI-SKMEPLEI
Chicken_CRY1_621aa      AAIKKLASEAGVEVIVRISHTLYDLDKIIELNGGQPPLTYKRFQT----LI-SRMEPLEM
Zebra_fish_CRY1_658aa   AAIKKLAMEAGVEVIVKTSHTLYNLDKIIELNGGQPPLTYKRFQT----LI-SRMDPPEM
Arabidopsis_CRY1_681aa  HRAKDVLTAQGIAVRSFNADLLYEPWEVTDEL-GRPFSMFAAFWERCLSMPYDPESPLLP
PutativeCow_CRY1_298aa  AAIKKLATEAGVEVIVRISHTLYDLDKIIELNGGQPPLTYKRFQT----LI-SKMEPLEI

Human_CRY1_586aa        PVETITSEVIEKCTTPLSDDHDEKYGVPSLEELGFDTDGLSSAVWPGGETEALTRLERHL
Mouse_CRY1_606aa        PADTITSDVIGKCMTPLSDDHDEKYGVPSLEELGFDTDGLSSAVWPGGETEALTRLERHL
Cow_CRY1_587aa          PVETITSEVIEKCTTPLSDDHDEKYGVPSLEELGFDTDGLPSAVWPGGETEALTRLERHL
Chicken_CRY1_621aa      PVETITPEVMQKCTTPVSDDHDEKYGVPSLEELGFDTDGLPSAVWPGGETEALTRLERHL
Zebra_fish_CRY1_658aa   PVETLSNSIMGCCVTPVSEDHGDKYGVPSLEELGFDIEGLPSAVWPGGETEALTRIERHL
Arabidopsis_CRY1_681aa  PKKIISGDVSKCVADPLVFEDDSEKGSNALLARA------WSPGWSNGDKALTTFINGPL
PutativeCow_CRY1_298aa  PVETITSEVIEKCTTPLSDDHDEKYGVPSLEELGFDTDGLPSAVWPGGETEALTRLERHL

Human_CRY1_586aa        ERKAWVANFERPRMNANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLY----KKVKKNSS
Mouse_CRY1_606aa        ERKAWVANFERPRMNANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLY----KKVKKNSS
Cow_CRY1_587aa          ERKAWVANFERPRMNANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLY----KKVKKNSS
Chicken_CRY1_621aa      ERKAWVANFERPRMNANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLY----KKVKKNSS
Zebra_fish_CRY1_658aa   ERKAWVANFERPRMNANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLY----RKVKKTST
Arabidopsis_CRY1_681aa  ------LEYSKNRRKA--DSATTSFLSPHLHFGEVSVRKVFHLVRIKQVAWANEGNEAGE
PutativeCow_CRY1_298aa  ERKAWVANFERPRMNANSLLASPTGLSPYLRFGCLSCRLFYFKLTDLY----KKVKKNSS

Human_CRY1_586aa        PPLSLYG-QLLWREFFYTAATNNPRFDKMEGNPICVQIPWDKNPEALAKWAEGRTGFPWI
Mouse_CRY1_606aa        PPLSLYG-QLLWREFFYTAATNNPRFDKMEGNPICVQIPWDKNPEALAKWAEGRTGFPWI
Cow_CRY1_587aa          PPLSLYG-QLLWREFFYTAATNNPRFDKMEGNPICVQIPWDKNPEALAKWAEGRTGFPWI
Chicken_CRY1_621aa      PPLSLYG-QLLWREFFYTAATNNPRFDKMEGNPICVQIPWDKNPEALAKWAEGRTGFPWI
Zebra_fish_CRY1_658aa   PPLSLYG-QLLWREFFYTAATTNPRFDKMEGNPICVRIPWDKNPEALAKWAEAKTGFPWI
Arabidopsis_CRY1_681aa  ESVNLFLKSIGLREYSRYISFNHPYSHERPLLGHLKFFPWAVDENYFKAWRQGRTGYPLV
PutativeCow_CRY1_298aa  PPLSLYG-QLLWREFFYT------------------------------------------
```

Q6.



Q7.

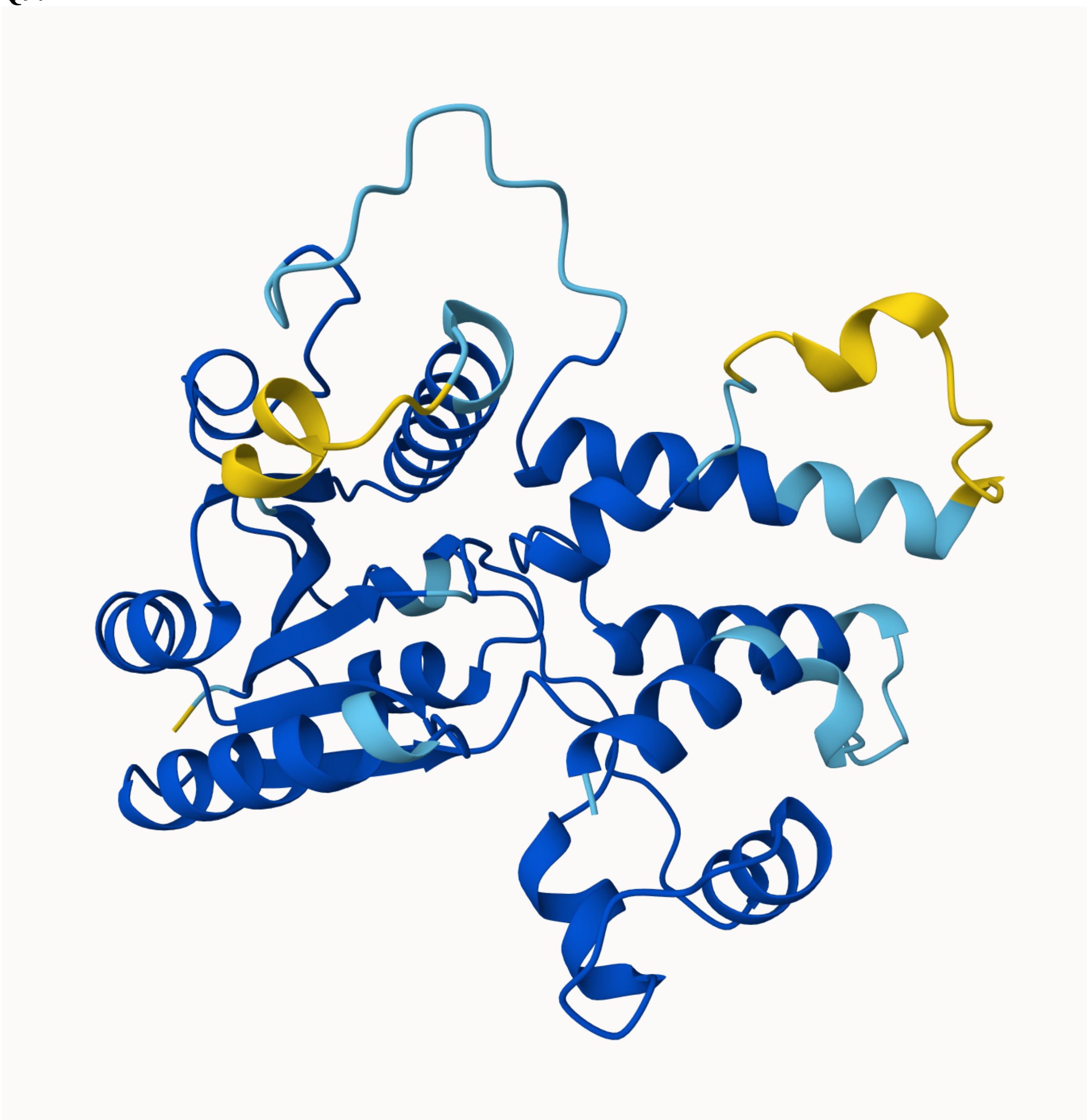


Q8.

I used Human CRY1, as the putative cow CRY1 has so many gaps. Below is the result:

PDB_ID <chr>	Method <chr>	Resolution <dbl>	Source <chr>	Evalue <dbl>	Identity <dbl>
4K0R	X-ray	2.65	Mus musculus	0	93.399
9DC0	X-ray	1.91	Erithacus rubecula	0	96.863
4CT0	X-ray	2.45	Mus musculus	0	98.000

Q9.



3D structural model of the Putative Cow CRY1 protein generated by AlphaFold. The model is colored by pLDDT confidence score.

Q10.

I performed a BLAST search of the ChEMBL database using the putative cow CRY1 sequence. The search identified Human Cryptochrome-1 (CHEMBL4296246) as a significant homolog

(99.3% identity, E-value 0). The human homolog has significant data reported. I identified 9 bioactivity records (mostly EC50 measurements) and 1 binding assay. Additionally, ligand efficiency data (Binding Efficiency Index vs. Surface Efficiency Index) was available. A key associated compound is KL001, a carbazole derivative identified as a Cryptochrome stabilizer that lengthens the circadian period.